Honors Biology Unit 3A Enzymes and ATP Script

Slides 17 through 27

17. We are now shifting from enzymes to cellular energy. Enzymes help reactions happen at the rates cells need, but cells also need a way to move energy from one reaction to another. ATP serves this purpose. As we study ATP, pay attention to its structure and how cells use it to power their activities.

18. ATP stands for adenosine triphosphate and is often described as the cell's energy-carrying molecule. It contains adenine, a ribose sugar, and three phosphate groups. Adenine and ribose form adenosine, while triphosphate means three phosphate groups. The phosphate end of the molecule transfers energy.

19. Cells use ATP to power activities such as active transport, muscle contraction, and the building of larger molecules. The three phosphate groups have negative charges that repel one another, making ATP relatively unstable. When ATP reacts with water, it forms more stable products. This reaction releases usable energy that the cell can connect to other processes.

20. This reaction is called ATP hydrolysis because water is used to remove ATP's third phosphate group. ATP becomes ADP, which has two phosphate groups, and inorganic phosphate is released separately. The overall reaction releases energy. Cells couple this energy-releasing reaction to energy-requiring processes, allowing ATP to power cellular work.

21. The phosphate removed from ATP is often transferred to another molecule. This process is called phosphorylation. Adding the phosphate can change the molecule's shape, increase its energy, or make it more reactive. For example, phosphorylation can activate a protein or prepare a reactant for the next step in a chemical reaction.

22. ATP is constantly recycled inside the cell. The energy released when food molecules are broken down is used to add a phosphate group back to ADP, reforming ATP. The cell can then break down ATP when energy is needed. The ATP-ADP cycle links energy-releasing reactions to cellular processes that require it.

23. This diagram shows how ATP moves through a repeating cycle. Energy is needed to add a phosphate to ADP and produce ATP. When ATP loses its third phosphate through hydrolysis, it releases energy and forms ADP again. ATP is not used for long-term energy storage. Instead, it acts as a rechargeable energy carrier that cells use continuously.

24. Organic molecules contain both stored chemical energy and materials that cells can reuse. Heterotrophs obtain these molecules by eating other organisms or their products. Digestion breaks food into smaller molecules, including simple sugars, amino acids, and fatty acids. Cells can use these molecules to build cellular structures or break them down to make ATP.

25. We will use glucose as our main example for studying how cells obtain energy from food. During cellular respiration, cells break down glucose through many small, controlled reactions. Releasing the energy in steps allows the cell to capture some of it in ATP. If the cell released all the energy at once, it would lose much more as heat.

26. Cells harvest energy from fuel by breaking large molecules into smaller ones and transferring electrons and hydrogen atoms between molecules. These electrons carry potential energy. As they move through a series of reactions, cells can capture their energy in new bonds and energy-carrying molecules. These controlled transfers are central to cellular respiration.

27. Oxidation and reduction describe the transfer of electrons between substances. Oxidation is the loss of electrons or hydrogen, while reduction is the gain of electrons or hydrogen. These reactions always occur together because one substance loses the electrons that another substance gains. The phrase OIL RIG can help you remember: oxidation is loss, and reduction is gain. Redox reactions help cells transfer energy from glucose to ATP in manageable steps.